



Laboratory Protocol

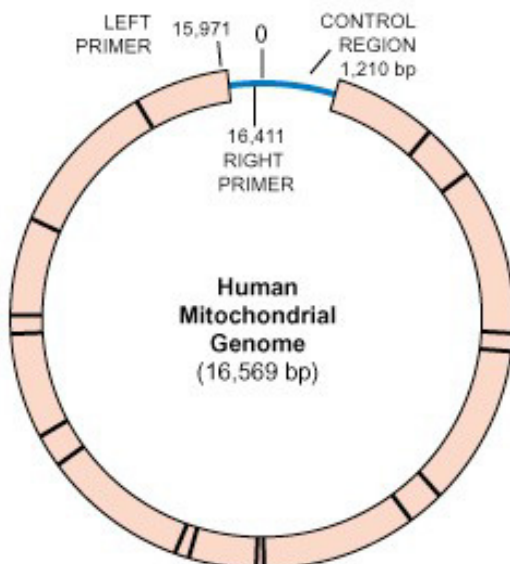
INTRODUCTION

Every human cell has a “second” genome found in the cell’s energy-generating organelle, the mitochondria. There are several hundred to several thousand mitochondria per cell. This means that the mitochondrial (mt) genome is highly amplified. While each cell contains only two copies of a given nuclear gene (one on each of the paired chromosomes), there are thousands of copies of a given mitochondrial gene per cell. Because of this high copy number, it is possible to obtain a mitochondrial DNA type from the equivalent of a single cell’s worth of mt DNA. Thus, mt DNA is the genetic system of choice in cases where tissue samples are very old, very small, or badly degraded by heat and humidity.

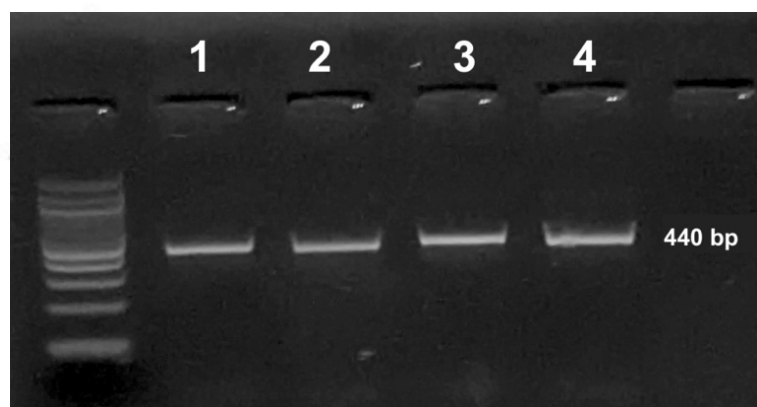
Under ideal circumstances—working from fresh cell samples—mitochondrial DNA is the easiest human DNA to amplify by PCR. This experiment examines a 440-nucleotide sequence from the noncoding region of the mt genome. Because each student is amplifying the same region, the gel electrophoresis results will be the same for each, with a band at 440 bp, however the specific mt DNA sequences will vary. Amplified student samples will be submitted to AZENTA. AZENTA will generate student mt DNA sequences and post the results to our Sequence Server. Instructions for accessing and analyzing DNA sequences will be provided.

Comparison of this small portion of the mt DNA genome reveals that most people have a unique pattern of single nucleotide polymorphisms (SNPs). These sequence differences, in turn, are the basis for far-ranging investigations on human DNA diversity and the evolution of hominids. Mitochondrial DNA is inherited from the egg cell, so genetic mutations only come from one branch on the family tree. This makes tracing lineage easier for mt DNA than nuclear DNA.

Mitochondrial Control Region



RESULTS OF GEL ELECTROPHORESIS





PROCEDURE

Part I: DNA Isolation by Swab

1. Label a tube containing 500 μ L 0.9% saline with a number, as assigned by your teacher. This way, your results will be anonymous. This number will be used throughout the protocol.
2. Using a sterile cotton swab with a 6" wood handle, thoroughly swab the inside of your cheek for 30 seconds, turning the swab occasionally so that the entire surface of the swab has been used.
3. Swirl the swab in the saline tube for 30 seconds, rotating the wood stick rapidly between your fingers so that the cheek cells fall off of the swab into the liquid.
4. Discard the swab into a special container designated for saliva waste.
5. Place your sample tube, together with other student samples, in a balanced configuration in a microcentrifuge, and spin for 1 minute.
6. Carefully pour off supernatant into the saliva waste container. Be careful not to disturb the cell pellet at the bottom of the test tube. A small amount of saline will remain in the tube.
7. Resuspend cells in remaining saline by flicking the base of the tube vigorously.
8. Use a micropipette to withdraw 30 μ L of cell suspension, and add cells to the tube containing 100 μ L of Chelex. Shake well to mix.
9. Boil cell sample for 10 minutes. Use boiling water bath, heat block, or program thermal cycler for 10 minutes at 99°C. Then, cool tube briefly on ice (optional).
10. After boiling, shake tube. Place in a balanced configuration in a microcentrifuge, and spin for 1 minute.
11. Use a micropipette to transfer 30 μ L of supernatant (containing the DNA) to clean 1.5 mL tube. *Avoid cell debris and Chelex beads.* This sample will be used for setting up the PCR reaction.
12. Store your sample on ice or in the freezer until ready to begin Part II.

Part II: DNA Amplification by PCR

1. Use a micropipette with a fresh tip to add 22.5 μ L of the appropriate primer mix to a PCR tube containing a Ready-To-Go PCR Bead. Tap the tube with your finger to dissolve the bead. Label the tube with your number.
2. Use a micropipette with a fresh tip to add 2.5 μ L of human DNA (from Part I) to the reaction tube, and tap to mix. Pool reagents by pulsing in a microcentrifuge or by sharply tapping tube bottom on lab bench.
3. Store all samples on ice until ready to amplify. Program thermal cycler for 30 cycles according to the following cycle profiles.
 - Denaturing time and temperature 30 sec - 94°C
 - Annealing time and temperature 30 sec - 58°C
 - Extending time and temperature 30 sec - 72°C
 - Hold at 10°C on thermocycler until PCR products can be moved to the refrigerator



Part III: DNA Analysis by Gel Electrophoresis

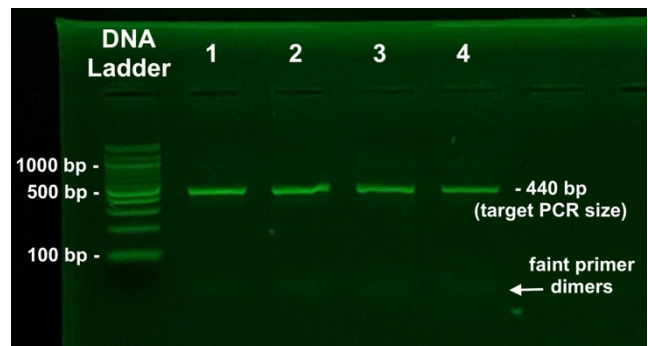
1. Use a micropipette with a fresh tip to add 6 μL of PCR sample into your assigned well of a 2% agarose gel. (This will leave enough product if you intend to sequence the mt control region.) Expel any air from the tip before loading, and be careful not to push the tip of the pipet through the bottom of the sample well.
2. Load 5 μL of the 100 bp size marker into the first lane of each gel.
3. Electrophorese at 140 volts for 20-30 minutes. Adequate separation will have occurred when the cresol red dye front has moved at least 2 cm from the wells.
4. Gels will contain DNA stain and be ready to view with a UV transilluminator immediately. DNA bands will be bright green.

RESULTS AND DISCUSSION

Interpreting Your Gel

1. Observe the photograph of the stained gel containing your sample and those from other students. Orient the photograph with the sample wells at the top. Interpret the band(s) in each lane of the gel:

- a. Scan across the photograph to get an impression of what you see in each lane. You should notice that virtually all student lanes contain one prominent band.
- b. Now locate the lane containing DNA Ladder. We buy the 100 bp ladder from NEB, and each line represents approximately 100 bp. You can find the reference photo at NEB online.



- c. The amplification product of 440 bp should be near the 500 bp marker (500 and 1000 bp bands will be brightest).
- d. It is common to see a second band lower on the gel. This diffuse (fuzzy) band is “primer dimer,” an artifact of the PCR reaction that results from the primers overlapping one another and amplifying themselves. Primer dimer is approximately 50 bp, and should be in a position ahead of the 121-bp marker.
- e. Additional faint bands, at other positions on the gel, occur when the primers bind to chromosomal loci other than mt control region and give rise to “nonspecific” amplification products. This is rare, but can lead to errors in sequencing results, if the sequencing primers bind to both pieces of DNA.



Thought Questions

1. How would you interpret a lane in which you observe primer dimer, but no 440-bp band?
2. The mt control region mutates at approximately ten times the rate of nuclear DNA. Propose a biological reason for the high mutation rate of mt DNA.
3. The high mutability of the mt genome means that it evolves more quickly than the nuclear genome. This makes the mt control region a laboratory for the study of DNA evolution. However, can you think of any drawbacks to this high mutation rate?
4. There are numerous insertions of mt DNA into nuclear chromosomes. Notably, scientists recently discovered a 540-bp fragment of the mt control region that inserted into chromosome 11 approximately 350,000 years ago. Would you expect any difference in the mutation rates of the control region sequence in the mt genome versus the chromosome 11 insertion? What implication does this have in the study of human evolution?

FURTHER SUGGESTIONS for exercises can be found at the DNA Learning Center's website Genetic Origins (<http://www.geneticorigins.org/geneticorigins/mito/mitoframeset.htm>).